# Supplementary Information for "Resistance-Aware Polypharmacology of Antimalarial Leads from African Natural Products: Docking, Network Scoring, and Targeted Molecular Dynamics"

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Table S1: Docking protocol validation: enrichment metrics and re-docking accuracy. Values reproduce the corrected benchmarks from the parent study (V2-corrected grids; AutoDock Vina + DiffDock consensus). MMV Malaria Box values are consensus-score ROC-AUCs for the three targets with MMV docking data (PfDHFR, PfCRT, PfATP4); DEKOIS 2.0 validation used AutoDock Vina only for PfDHFR. BEDROC values use $\alpha = 20$ (MMV). Re-docking RMSD is the heavy-atom RMSD between the top-scoring docked pose and the co-crystallised ligand; MTX redocking failed (RMSD $\approx$ 30 Å).

| Target | Method | ROC-AUC | EF1 % | EF5 % | BEDROC | RMSD (Å) |
|---|---|---|---|---|---|---|
| PfDHFR (7F3Y) | Vina redock | – | – | – | – | N/A |
| | MMV consensus (Vina+DiffDock) | 0.924 | – | 2.57 | 1.309 | – |
| | DEKOIS (Vina only) | 0.450 | – | 0.00 | – | – |
| PfCRT (6UKJ) | MMV consensus (Vina+DiffDock) | 0.971 | – | 1.11 | 9.434 | – |
| PfATP4 (9N10) | MMV consensus (Vina+DiffDock) | 1.000 | – | 2.00 | 1.810 | – |
| PfClpR (4GM2; not PfClpP) | MMV enrichment | not evaluated | – | – | – | – |

**Identity note.** PDB 4GM2 is PfClpR rather than PfClpP; it is not used as PfClpP validation.

Table S2: Endpoint-analysis interpretation for the four parent-study consensus complexes used for production MD. These systems are distinct from the 17 set-C polypharmacology candidates. Mean ± standard deviation is shown only for the single interpretable MM-GBSA result from 101 snapshots; dissociated or conversion-corrupted systems are reported as N/A. Mutant RRS is a docking-based analysis of set C and is not assigned to these MD systems.

| Ligand | Target | $\Delta G_{\mathrm{bind}}$ | Std. dev. | Interpretation |
|---|---|---|---|---|
| 164 | PfClpR-labelled cohort*-WT | – | – | N/A; ligand dissociated‡ |
| 201 | PfDHFR-WT | – | – | N/A; ligand dissociated‡ |
| 214 | PfCRT-WT | −18.25 | 0.40 | Interpretable (gmx_MMPBSA) |
| 438 | PfATP4-WT | – | – | N/A; conversion corrupted§ |

*PDB 4GM2 is PfClpR rather than PfClpP; the 164-labelled system is therefore not structural validation of PfClpP.
‡Ligand dissociated during production; no binding free energy is inferred.
§Excluded due to two-chain topology incompatibility with CHARMM36-to-AMBER conversion.

*Scope of the MD evidence.* The parent-study trajectories provide a targeted structural check on four wild-type complexes; they do not constitute candidate-specific Set-C production MD. A 16-system Set-C MD pilot (PP-01 and PP-02 against the six-mutant panel, 10 ns each) passed the

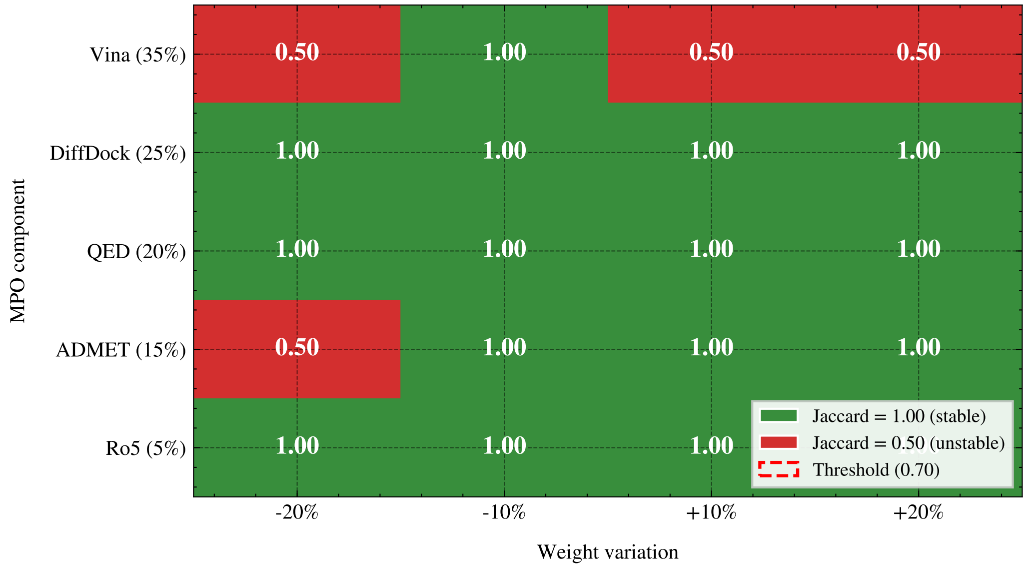


Figure S1: MPO sensitivity analysis heatmap. Each cell shows the Jaccard similarity of the top-20 hit list when the corresponding MPO weight is varied by ±20 %. Green (1.00) indicates the hit list is unchanged; red (0.50) indicates substantial reordering.

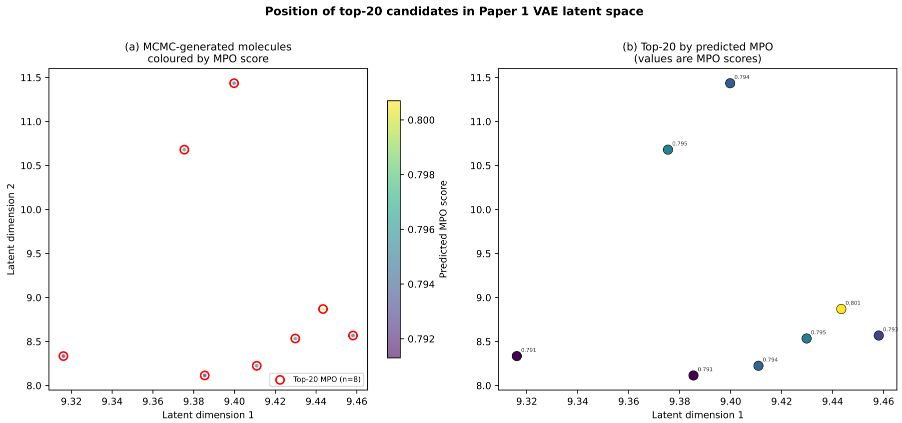


Figure S2: Position of the top-20 candidates in the Paper 1 VAE latent space. Molecules are coloured by predicted MPO score; top-20 by MPO are highlighted in red.

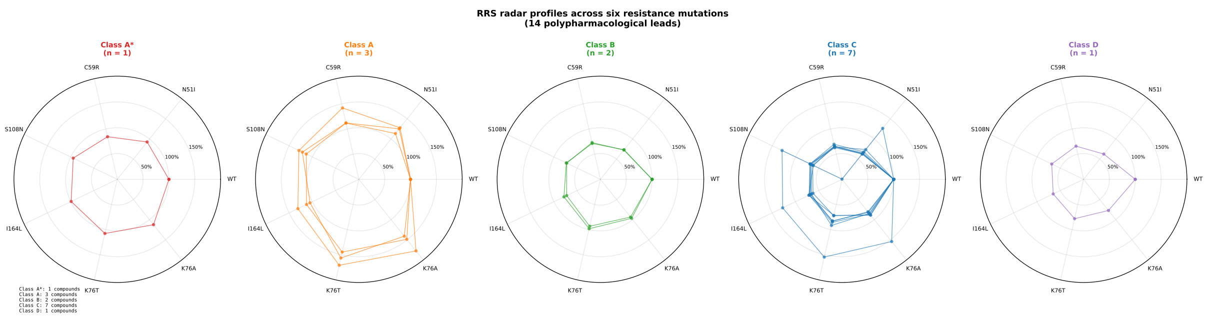


Figure S3: Per-compound, docking-derived RRS profiles across all six resistance mutations for the 17 set-C polypharmacological leads. The four parent-study MD systems are not included. Each axis represents one mutation; concentric rings indicate 50 %, 100 %, and 150 % RRS (WT-normalised). Larger radial values indicate greater preservation of the docking score relative to the corresponding wild-type target; the plot does not encode absolute binding potency.

Table S3: Predicted ADMET profile of the 17 set-C polypharmacology candidates (ADMET-AI, parent-library screening). All values are machine-learning predictions; experimental ADMET profiling and three-tool concordance were outside the scope of this study. Range across candidates: LogS $[0.34, 0.73]$, CYP3A4 inhibition $\geq 0.5$ for 7/17 candidates, Caco-2 permeability $\geq 0.5$ for 15/17, Clint$_h$ $[0.48, 0.63]$.

| Compound | ADMET-AI LogS | CYP3A4 prob. | Caco-2 prob. | Clint$_h$ |
|---|---|---|---|---|
| PP-01 | 0.34 | 0.39 | 0.63 | 0.60 |
| PP-02 | 0.46 | 0.97 | 0.86 | 0.55 |
| PP-03 | 0.43 | 0.62 | 0.95 | 0.55 |
| PP-04 | 0.40 | 0.58 | 0.92 | 0.63 |
| PP-05 | 0.61 | 1.00 | 0.35 | 0.51 |
| PP-06 | 0.54 | 0.90 | 0.97 | 0.60 |
| PP-07 | 0.36 | 0.02 | 0.75 | 0.62 |
| PP-08 | 0.59 | 1.00 | 0.41 | 0.56 |
| PP-09 | 0.65 | 0.13 | 0.81 | 0.57 |
| PP-10 | 0.49 | 0.13 | 0.93 | 0.62 |
| PP-11 | 0.39 | 0.06 | 0.96 | 0.58 |
| PP-12 | 0.49 | 0.00 | 0.98 | 0.48 |
| PP-13 | 0.57 | 0.97 | 0.98 | 0.53 |
| PP-14 | 0.66 | 0.05 | 0.68 | 0.49 |
| PP-15 | 0.66 | 0.49 | 0.95 | 0.51 |
| PP-16 | 0.54 | 0.25 | 0.91 | 0.50 |
| PP-17 | 0.73 | 0.01 | 0.82 | 0.50 |

Table S4: ACSI weight sensitivity analysis. Each of the four ACSI weights ($w_{\mathrm{DrugBank}}$, $w_{\mathrm{ANPDB}}$, $w_{\mathrm{fsp^3}}$, $w_{\mathrm{NPL}}$) was varied by $\pm 20\,\%$ while keeping the remaining weights proportional. The Jaccard index measures overlap of the resulting top-20 ACSI-ranked list with the baseline ranking. A Jaccard $\geq 0.70$ (14/20 compounds stable) indicates robustness.

| Weight varied | Variation (%) | Mean ACSI (top-20) | Std. dev. | Jaccard vs. baseline |
|---|---|---|---|---|
| *Data to be populated after ACSI computation.* | | | | |

prespecified trajectory-QC gate (16/16) and is reported as a secondary analysis; its trajectory-derived MD-RRS and mutant-state MM-GBSA estimates are summarised in Table S7 and compared with docking RRS in Figure S4.

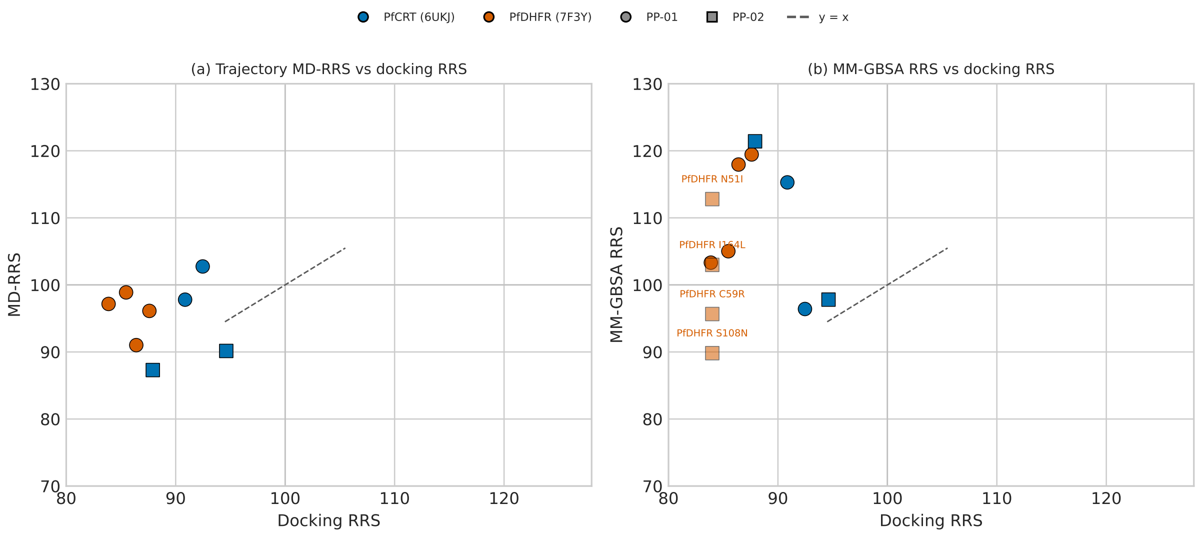


Figure S4: Set-C MD pilot trajectory-derived RRS estimates versus docking RRS (WT-normalised, WT = 100). Panel (a): MD-RRS (mean-minimum-distance geometry) versus docking RRS for the eight mutant systems with both estimates; panel (b): MM-GBSA RRS versus docking RRS, with the four PP-02 PfDHFR systems (no docking WT reference in the pilot) shown on the MM-GBSA axis only. Points are coloured by target (PfCRT, blue; PfDHFR, orange) and shaped by candidate (PP-01, circle; PP-02, square). Docking predicted looser mutant binding for all systems (RRS $< 100$); the trajectory-derived estimates do not reproduce this pattern. Secondary, exploratory analysis; no primary claim is based on it.

The stable identifiers were retrieved from PlasmoDB/VEuPathDB annotations on 12 August 2026 (PlasmoDB Consortium, 2026); record-level permalinks and the retrieval date are preserved in the machine-readable CSV. The exact database release/build was not captured in the source snapshot and is therefore not treated as an inferred annotation version. Because the present study does not provide an independent multi-gene targetome or a prespecified parasite-proteome background, no GO or pathway-enrichment test was performed. This table therefore supplies biological identity and mutation provenance without converting docking scores into pathway-level evidence (Trapotsi et al., 2022).

## References

PlasmoDB Consortium. PlasmoDB: The *Plasmodium* genomic resource. https://plasmodb.org/, 2026.

Maria-Anna Trapotsi, Layla Hosseini-Gerami, and Andreas Bender. Computational analyses of mechanism of action (moa): data, methods and integration. *RSC Chemical Biology*, 3(2): 170–200, 2022. doi: 10.1039/D1CB00069A.

## Use of Artificial Intelligence

During the preparation of this work the authors used AI assistants to support code development and data analysis scripting. The authors reviewed and edited all AI-assisted output and take

Table S5: Stable *Plasmodium falciparum* target identifiers and mutation context used in this study. Identifiers are reported for annotation and provenance only; they do not constitute pathway-level target engagement, causal network evidence, or experimental validation. The accompanying source-data record preserves the target annotations and retrieval provenance.

| Target | Stable gene ID | Structural context | Mutation or evidence boundary |
|---|---|---|---|
| PfDHFR | `PF3D7_0417200` | 7F3Y WT template | N51I, C59R, S108N, I164L; antifolate-resistance states |
| PfCRT | `PF3D7_0709000` | 6UKJ WT template | K76T clinical resistance marker; K76A comparator state |
| PfATP4 | `PF3D7_1211900` | 9N10 parent-study structure | Target annotation only; not part of the Set-C mutation panel |
| PfClpP | `PF3D7_0307400` | Proteolytic Clp subunit | Not assigned to PDB 4GM2; no PfClpP claim is inferred from the 4GM2 record |
| PfClpR | `PF3D7_1436800` | 4GM2, PfClpR(P61–E244) | 4GM2 is PfClpR, not PfClpP; parent-study identity retained explicitly |

Table S6: Cohort and estimand separation. The Set-C docking cohort and the parent-study MD cohort are non-overlapping. The Set-C MD pilot passed the prespecified trajectory-QC gate (16/16) and is reported as a secondary, exploratory analysis whose MD-RRS and MM-GBSA estimates are not merged into the primary docking-RRS classification.

| Evidence layer | Cohort | Systems | Primary analyses | Interpretation boundary |
|---|---|---|---|---|
| Set-C docking | PP-01–PP-17 | 136 | WT/mutant docking; RRS, ACSI, PNS | Docking-derived prioritisation; not measured binding or MD-RRS |
| Parent-study MD | Ligands 164, 201, 214, 438 | 4 WT complexes × 10 ns | RMSD, contacts, bound-state assessment, MM-GBSA where valid | Separate structural check; one interpretable MM-GBSA estimate (PfCRT–214) |
| Set-C MD pilot | PP-01/PP-02 | 16 systems (16/16 QC PASS) | Trajectory QC, discriminative MD-RRS, MM-GBSA | Reported as secondary; not merged into primary RRS classification |
| Full Set-C MD-RRS | PP-01–PP-17 | 136 planned systems | Full trajectory QC and MD-RRS | Not computed; requires complete PASS-QC cohort |

Table S7: Set-C MD pilot endpoint estimates (secondary, exploratory analysis). All 16 systems passed the prespecified trajectory-QC gate (16/16); $\Delta G_{\text{bind}}$ is the mean over 100 snapshots $\pm$ standard deviation. MMG-RRS and MD-RRS use the same mutant/WT ratio convention as the docking RRS (WT = 100); n/a = no WT reference available in the pilot for PP-02 PfDHFR (docking RRS was computed against the parent 17-candidate cohort only). The two systems flagged with * failed the first MM-GBSA pass (BOND/UB overflow from a protein split across the periodic boundary) and were re-run on PBC-whole trajectories.

| Candidate | Target | State | $\Delta G_{\text{bind}}$ (kcal mol$^{-1}$) | SD | MMG-RRS | MD-RRS | Docking RRS |
|---|---|---|---|---|---|---|---|
| PP-01 | PfCRT | K76A | −35.29 | 1.17 | 115.3 | 97.8 | 90.86 |
| PP-01 | PfCRT | K76T | −29.51 | 2.65 | 96.4 | 102.8 | 92.47 |
| PP-01 | PfCRT | WT | −30.61 | 1.65 | 100.0 | 100.0 | – |
| PP-01 | PfDHFR | C59R | −26.56 | 1.91 | 105.0 | 98.9 | 85.47 |
| PP-01 | PfDHFR | I164L | −26.13 | 2.34 | 103.3 | 97.2 | 83.87 |
| PP-01 | PfDHFR | N51I | −29.83 | 1.50 | 118.0 | 91.0 | 86.40 |
| PP-01 | PfDHFR | S108N | −30.21 | 3.62 | 119.5 | 96.1 | 87.60 |
| PP-01 | PfDHFR | WT | −25.29 | 2.94 | 100.0 | 100.0 | – |
| PP-02 | PfCRT | K76A* | −24.53 | 1.27 | 97.8 | 90.2 | 94.62 |
| PP-02 | PfCRT | K76T | −30.45 | 1.38 | 121.4 | 87.3 | 87.91 |
| PP-02 | PfCRT | WT | −25.08 | 2.56 | 100.0 | 100.0 | – |
| PP-02 | PfDHFR | C59R | −28.85 | 1.88 | 95.7 | 102.0 | n/a |
| PP-02 | PfDHFR | I164L | −31.06 | 1.52 | 103.0 | 97.4 | n/a |
| PP-02 | PfDHFR | N51I | −34.02 | 2.68 | 112.8 | 72.2 | n/a |
| PP-02 | PfDHFR | S108N | −27.09 | 1.55 | 89.8 | 97.5 | n/a |
| PP-02 | PfDHFR | WT* | −30.16 | 2.06 | 100.0 | 100.0 | – |

full responsibility for the content of this article. No generative AI tool is listed as an author. All computational results, statistical analyses, and quantitative claims were generated and verified by the authors.